Designing the Research Plan

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What Is a Research Plan?

A research plan is a practical map that explains how a study will be carried out. ➢

It connects the research question with the design, participants or samples, measurements, ➢ analysis, ethics, and time.

It should be prepared before data collection begins. ➢

Why Do We Need a Research Plan?

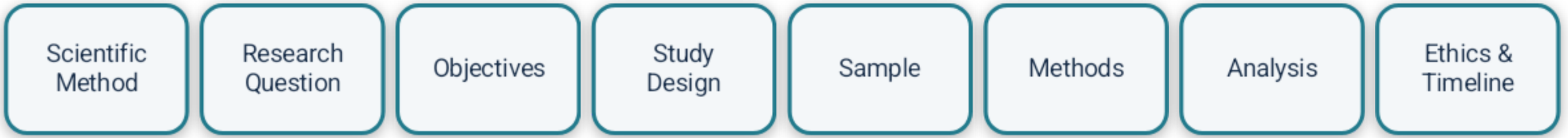
Clarity — turns an idea into organized steps.

Feasibility — shows whether the study can actually be completed.

Consistency — keeps procedures similar for all participants or samples.

Quality — reduces avoidable errors, bias, and missing information.

From Scientific Method to Research Plan



Scientific Method explains how we think scientifically. The Research Plan converts that logic into practical steps.

The Main Question of This Lecture

I have a research question. How do I turn it into a study that can actually be performed?

A good plan tells us what we will study, who we will study, what we will measure, and how we will analyse the evidence.

Step 1 — Define the Research Question

A good research question is clear, focused, measurable, and feasible.

Example: Is serum ferritin associated with hemoglobin level in hemodialysis patients? ➢

The question guides the rest of the research plan.

Step 2 — Set the Research Objectives

General objective (Aim) — states the main purpose of the study. ➢

Specific objectives — divide the main purpose into measurable tasks. ➢

Example: Measure ferritin and hemoglobin, then assess their association.

Research Question vs. Objective

Research question — asks what we want to know. ➢

Objective — states what we will do to answer the question. ➢

Question: Is ferritin associated with haemoglobin?

Objective: To assess the association between serum ferritin and haemoglobin level.

Step 3 — Choose the Study Design

The study design is the basic structure used to obtain evidence. ➢

At this stage, we choose a design that matches the research question. ➢

Two broad approaches are ➢

Observational studies. ➢

Experimental studies. ➢

Observational Study Designs

Three common designs — illustrated with anemia and iron status

01 Cross-sectional Study

Measure exposure and outcome at the same time

Exposure
Ferritin level

Outcome
Hemoglobin level

Measure both in 100 patients at one time

02 Case-control Study

Start with the outcome → Examine previous exposure

Cases
50 patients with anemia

Controls
50 people without anemia

Compare previous iron deficiency

03 Cohort Study

Start with exposure → Observe new outcomes

Exposed
50 people with low ferritin

Unexposed
50 people with normal ferritin

Follow for 1 year → Who develops anemia?

Key idea: The researcher observes exposures and outcomes without assigning the exposure.

Experimental Study Design

Experimental study — the researcher assigns an intervention and compares outcomes. ▪

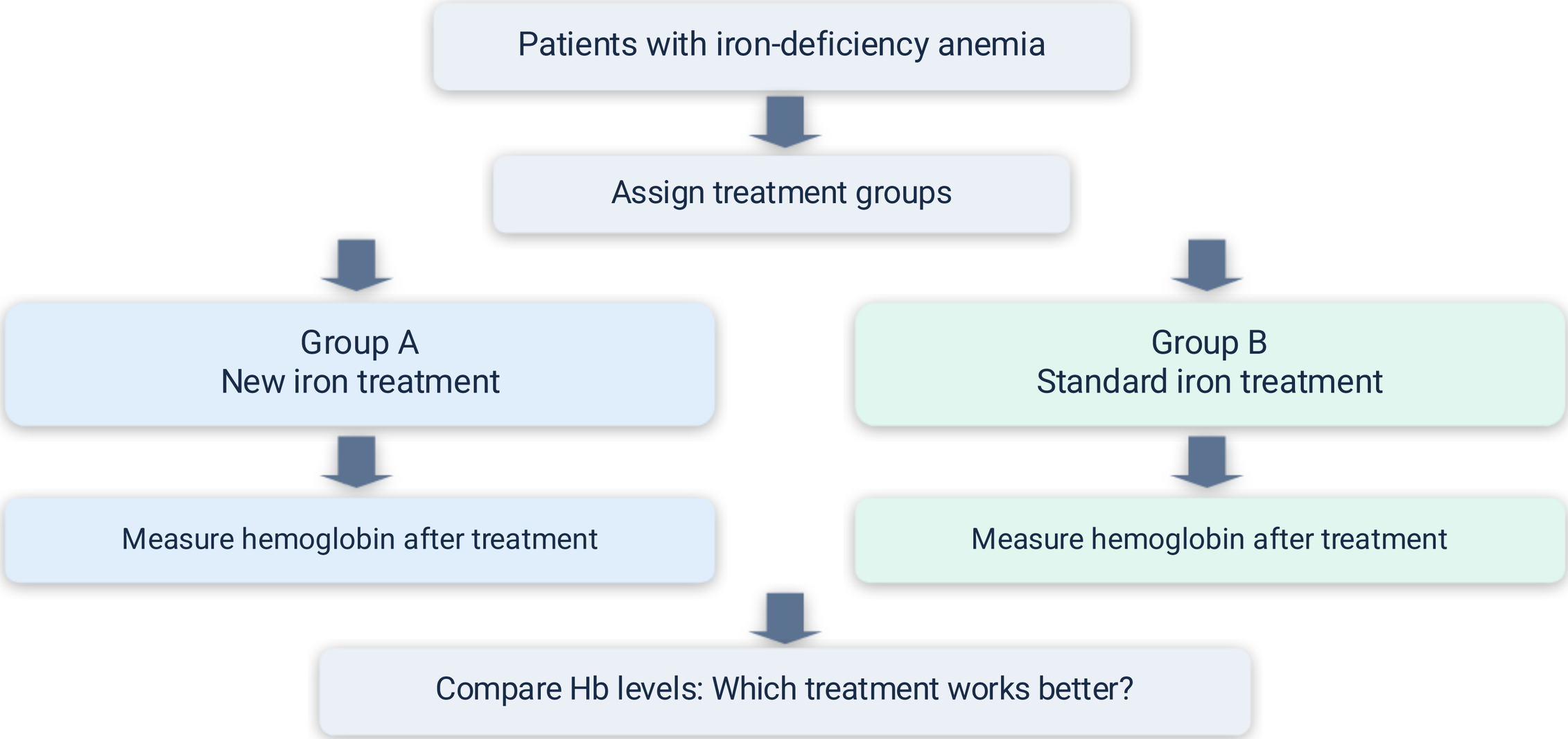
Clinical trial — an experimental study in human participants used to evaluate an intervention. ▪

Example: comparing a new diagnostic or treatment strategy with an appropriate comparison. ➢

Clinical trials will be discussed in detail in a later lecture.

Experimental Study Design

Example: Comparing two treatments for iron-deficiency anemia



Key idea: The researcher assigns an intervention and compares outcomes.

Observational vs. Experimental Study

Observational Study	Experimental Study
Researcher observes	Researcher assigns an intervention
No intervention is assigned	An intervention is assigned
Measures existing exposures and outcomes	Compares outcomes after intervention
Example: Measure Hb and ferritin levels	Example: Give iron treatment and measure Hb

Key difference: Does the researcher assign the intervention?

Step 4 — Define the Study Population

Target population — the wider group the study aims to understand.

Study population — the accessible group from which participants are recruited.

Sample — the participants or specimens actually included in the study.

Step 5 — Inclusion Criteria

Inclusion criteria define who is eligible to enter the study. ➢

They should be related to the research question and defined before recruitment. ➢

Example for a hemodialysis study: adults ≥18 years, receiving haemodialysis, and meeting the required dialysis duration. ➢

Clear criteria make participant selection consistent. ➢

Step 5 — Exclusion Criteria

- Exclusion criteria identify conditions that make an otherwise relevant participant unsuitable for this particular study.
- Examples may include an acute condition, recent treatment that affects the studied marker, inadequate specimen, or another predefined reason.
- Exclusion should be scientifically justified—not based on whether the result supports the hypothesis.

Step 6 — Sampling and Sample Size

Sampling method — how participants or specimens are selected. ➢

Sample size — the number of participants or specimens required for the study. ➢

A sample that is too small may give imprecise estimates or miss an important difference. ➢

A very large sample is not automatically better; the number should be scientifically justified ➢ and feasible.

What Determines Sample Size?

Sample size depends on five main factors:

1. **Study Aim and Design**
   What are we studying, and how?
2. **Expected Difference and Variation**
   Smaller differences or greater variation often need more samples.
3. **Required Precision**
   Higher precision usually needs more samples.
4. **Missing Data and Dropouts**
   Plan extra participants for incomplete data or loss to follow-up.
5. **Available Resources**
   Consider time, cost, participants, and laboratory materials.

KEY POINT • Calculate and justify sample size before starting the study.

Step 7 — Define the Variables

Exposure / independent variable — the factor being investigated. ➢

Outcome / dependent variable — the result being measured. ➢

Other relevant variables — may include age, sex, clinical factors, or potential confounders. ➢

Important variables need clear definitions and measurement methods. ➢

Plan Data and Specimen Collection

01
Plan Data Collection
Decide what data to collect and when

e.g., Age, sex, Hb level

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02
Collect Specimens
Follow a standard collection procedure

e.g., Venous blood sample

↓

03
Handle and Test Samples
Transport, process, store, and measure

e.g., Serum separation for ELISA

↓

04
Record Data
Use coded IDs and structured forms

e.g., Sample ID: S001

↓

05
Document Problems
Record missing data and deviations

e.g., Missing Hb result

Key idea: Standard procedures and accurate records improve data quality.

Plan Laboratory Quality

Four essential practices for reliable laboratory results

01 Validated / Verified Methods

Use methods proven suitable for the test.

Example: Verify ELISA kit performance

02 Calibration & Quality Control

Check calibration and test performance.

Example: Standards + control samples

03 Consistent Procedures

Use the same procedure for every sample.

Example: Same incubation time and temperature

04 Traceability

Record materials, equipment, and results.

Example: Sample ID, kit lot, test date

✓ Reliable and Consistent Laboratory Results

Step 10 — Plan the Data Analysis

- Decide how each objective will be analyzed before examining the final results.
- Plan descriptive statistics and appropriate comparisons or associations.
- Consider effect size, variability, uncertainty, and potential confounding.
- The analysis should answer the research question—not simply produce a p-value.

Step 11 — Ethics and Data Protection

Four essential principles for responsible medical research

Research Ethics & Data Protection

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01 Ethical Approval

Obtain required ethics approval before starting.

Example: Research ethics committee approval

02 Informed Consent

Explain the study and obtain valid consent.

Example: Signed participant consent form

03 Confidentiality

Protect identities and personal information.

Example: Use sample ID S001 instead of names

04 Patient Safety

Avoid unauthorized changes to patient care.

Example: Do not stop treatment for research

↓

Protect Participants' Rights, Privacy & Safety

Step 12 — Feasibility and Timeline

- **Feasibility — Can the study realistically be completed with the available participants, tests, equipment, staff, budget, and time?**
- **Timeline — When will recruitment, laboratory work, analysis, and reporting occur?**

Different Forms of Research — A Preview

- **Review-based research — uses existing published evidence and does not collect new participant or laboratory data.**
- **Original research — collects new data, such as laboratory, clinical, observational, experimental, or survey data.**

The detailed classification of research will be covered in a later lecture.

From the Research Plan to the Final Research

BEFORE THE STUDY — Research Plan

- Question & objectives ✓
- Study design ✓
- Population & sample ✓
- Variables & methods ✓
- Analysis plan ✓
- Ethics & timeline ✓

AFTER CONDUCTING THE STUDY — Final Research

- Abstract / Summary ➢
- Introduction ➢
- Literature Review ➢
- Materials and Methods ➢
- Results ➢
- Discussion ➢
- Conclusion & Recommendations ➢

Plan → Conduct → Analyse → Report

Main Parts of a Completed Research

Abstract / Summary — a brief overview of the whole study. ➢

Introduction — explains the problem and why it matters. ➢

Literature Review — summarizes what is already known and identifies the gap. ➢

Materials and Methods — explains how the study was performed. ➢

Results — reports what was found. ➢

Discussion — explains what the findings mean and compares them with previous evidence. ➢

Conclusion & Recommendations — states the main conclusion and appropriate future or practical ➢ recommendations.

References — lists the sources used. ➢

AI in Designing a Research Plan

Literature planning — suggest search terms and help organize topics.

Plan organization — help structure objectives, variables, checklists, or timelines.

Consistency checks — identify unclear definitions or missing links between question, variables, and analysis.

Writing support — improve clarity of a draft protocol or data-collection form.

AI Tools for Research Planning

Practical tools students can start using for free

Elicit

LITERATURE SEARCH

Find papers and extract study details.

Free plan • Paid upgrades

ResearchRabbit



RESEARCH MAPPING

Discover related studies and citation networks.

Free plan • Paid upgrades

ChatGPT

RESEARCH ORGANIZATION

Draft objectives, variables, and study outlines.

Free plan • Paid upgrades

Consensus



EVIDENCE SEARCH

Explore published evidence on a research question.

Free plan • Paid upgrades

Remember: AI supports research; always verify references and protect patient data.

AI Tools for Scientific Research

Two useful tools for literature review and research planning

Claude

Academic reading & writing

Summarize research papers, analyze documents, and help organize research objectives.

Example: Draft objectives for a study on anemia in CKD.

LeapSpace (Elsevier)

Scientific literature & evidence

Explore scientific literature, compare published findings, and investigate research gaps.

Example: Compare previous studies on anemia biomarkers.

Researcher responsibility: Verify citations, check accuracy, and protect patient data.

Putting the Plan Together — Laboratory Example

Question: Is serum ferritin associated with haemoglobin in haemodialysis patients?

Objectives: Measure ferritin and haemoglobin and assess their association.

Design: Cross-sectional observational study.

Population: Eligible haemodialysis patients; inclusion and exclusion criteria are defined in advance.

Measurements: Ferritin, haemoglobin, and relevant clinical variables.

Analysis: Describe the data and assess the planned association while considering relevant confounders.

Ethics: Approval, coded data, and confidentiality.

Summary

A research plan converts a scientific question into an organized and feasible study. ❖

Question → Objectives → Design → Population/Sample → Variables → Data Collection → Analysis → Ethics → Timeline ❖

Study-design types and research classification will be explored further in later lectures. ❖

The final research report presents what was planned, performed, found, and interpreted. ❖

AI can support planning, but scientific responsibility remains with the researcher. ❖

Thank you